
User Manual

for MPC560XB PORT Driver

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Rev 1.2





Contents

Section number	Title	Page
Chapter 1		
Revision History		
Chapter 2		
Introduction		
2.1	Supported Derivatives.....	9
2.2	Overview.....	10
2.3	About this Manual.....	10
2.4	Acronyms and Definitions.....	11
2.5	Reference List.....	11
Chapter 3		
Driver		
3.1	Requirements.....	13
3.2	Driver Design Summary.....	13
3.3	Deviation from Requirements.....	14
3.4	Software specification.....	15
3.4.1	Define Reference.....	15
3.4.1.1	Define PORT_E_DIRECTION_UNCHANGEABLE.....	15
3.4.1.2	Define PORT_E_INVALID_DIRECTION.....	15
3.4.1.3	Define PORT_E_MODE_UNCHANGEABLE.....	16
3.4.1.4	Define PORT_E_PARAM_CONFIG.....	16
3.4.1.5	Define PORT_E_PARAM_INVALID_MODE.....	16
3.4.1.6	Define PORT_E_PARAM_PIN.....	17
3.4.1.7	Define PORT_E_PARAM_POINTER.....	17
3.4.1.8	Define PORT_E_UNINIT.....	17
3.4.1.9	Define PORT_GETVERSIONINFO_ID.....	18
3.4.1.10	Define PORT_INIT_ID.....	18
3.4.1.11	Define PORT_REFRESHPINDIRECTION_ID.....	18
3.4.1.12	Define PORT_SETPINDIRECTION_ID.....	19

Section number	Title	Page
3.4.1.13	Define PORT_SETPINMODE_ID.....	19
3.4.2	Macros Reference.....	19
3.4.2.1	Macro SIUL_GPDI.....	20
3.4.2.2	Macro SIUL_GPDO.....	20
3.4.2.3	Macro SIUL_IFMC.....	20
3.4.2.4	Macro SIUL_MPGPDO.....	20
3.4.2.5	Macro SIUL_PCR.....	20
3.4.2.6	Macro SIUL_PGPDI.....	20
3.4.2.7	Macro SIUL_PGPDO.....	20
3.4.2.8	Macro SIUL_PSMI.....	20
3.4.3	Enum Reference.....	21
3.4.4	Function Reference.....	21
3.4.4.1	Function FUNC.....	21
3.4.5	Structs Reference.....	22
3.4.5.1	Structure Port_LLD_PinConfigType.....	22
3.4.5.2	Structure Port_LLD_UnUsedPinConfigType.....	23
3.4.6	Types Reference.....	24
3.4.7	Variables Reference.....	24
3.4.8	Port_SetPinMode function parameters.....	25
3.5	Symbolic Names Disclaimer.....	44

Chapter 4 Tresos Configuration Plug-in

4.1	Configuration elements of Port.....	45
4.2	Form IMPLEMENTATION_CONFIG_VARIANT.....	45
4.3	Form PortGeneral.....	46
4.3.1	PortDevErrorDetect (PortGeneral).....	47
4.3.2	PortSetPinDirectionApi (PortGeneral).....	47
4.3.3	PortSetPinModeApi (PortGeneral).....	48
4.3.4	PortVersionInfoApi (PortGeneral).....	48

Section number	Title	Page
4.3.5	PortPinAbstractedMode (PortGeneral).....	48
4.4	Form CommonPublishedInformation.....	49
4.4.1	ArReleaseMajorVersion (CommonPublishedInformation).....	49
4.4.2	ArReleaseMinorVersion (CommonPublishedInformation).....	50
4.4.3	ArReleaseRevisionVersion (CommonPublishedInformation).....	50
4.4.4	ModuleId (CommonPublishedInformation).....	51
4.4.5	SwMajorVersion (CommonPublishedInformation).....	51
4.4.6	SwMinorVersion (CommonPublishedInformation).....	52
4.4.7	SwPatchVersion (CommonPublishedInformation).....	52
4.4.8	VendorApiInfix (CommonPublishedInformation).....	53
4.4.9	VendorId (CommonPublishedInformation).....	53
4.5	Form PortConfigSet.....	54
4.5.1	Form UnUsedPortPin.....	54
4.5.1.1	PortPinInitialMode (UnUsedPortPin).....	54
4.5.1.2	PortPinWpe (UnUsedPortPin).....	55
4.5.1.3	PortPinReadback (UnUsedPortPin).....	55
4.5.1.4	PortPinSafeMode (UnUsedPortPin).....	56
4.5.1.5	PortPinWps (UnUsedPortPin).....	56
4.5.1.6	PortPinDirection (UnUsedPortPin).....	57
4.5.1.7	PortPinLevelValue (UnUsedPortPin).....	57
4.5.2	Form PortContainer.....	58
4.5.2.1	PortNumberOfPortPins (PortContainer).....	58
4.5.2.2	Form PortPin.....	58
4.5.2.2.1	PortPinWpe (PortPin).....	59
4.5.2.2.2	PortPinWps (PortPin).....	59
4.5.2.2.3	PortPinOde (PortPin).....	60
4.5.2.2.4	PortPinSafeMode (PortPin).....	60
4.5.2.2.5	PortPinDirectionChangeable (PortPin).....	60
4.5.2.2.6	PortPinReadback (PortPin).....	61

Section number	Title	Page
4.5.2.2.7	PortPinModeChangeable (PortPin).....	61
4.5.2.2.8	PortPinId (PortPin).....	61
4.5.2.2.9	PortPinPcr (PortPin).....	62
4.5.2.2.10	PortPinDirection (PortPin).....	62
4.5.2.2.11	PortPinInitialMode (PortPin).....	63
4.5.2.2.12	PortPinMode (PortPin).....	63
4.5.2.2.13	PortPinLevelValue (PortPin).....	64
4.5.2.2.14	PortPinSlewRate (PortPin).....	64

Chapter 1

Revision History

Table 1-1. Revision History

Revision	Date	Author	Description
1.0	08-Feb-2012	Alfredo Di Martino	First draft for BETA 0.9.0
1.1	13-May-2013	Tilak Basavaraju	Updated for RTM 1.0.0
1.2	02-June-2014	François Penalosa	Updated for RTM 1.0.1



Chapter 2

Introduction

This User Manual describes Freescale Semiconductor AUTOSAR Port (Port) for MPC5607B .

AUTOSAR Port driver configuration parameters and deviations from the specification are described in Port Driver chapter of this document. AUTOSAR Port driver requirements and APIs are described in the AUTOSAR Port driver software specification document.

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of Freescale Semiconductor .

Table 2-1. MPC560XB Derivatives

Freescale Semiconductor	mpc5601dxlh_lqfp64 mpc5601dxll_lqfp100 mpc5602dxlh_lqfp64 mpc5602dxll_lqfp100 mpc5602bxlh_lqfp64 mpc5602bxll_lqfp100 mpc5602bxlq_lqfp144 mpc5602cxlh_lqfp64 mpc5602cxll_lqfp100 mpc5603bxlh_lqfp64 mpc5603bxll_lqfp100 mpc5603bxlq_lqfp144 mpc5603cxlh_lqfp64 mpc5603cxll_lqfp100 mpc5604bxlh_lqfp64 mpc5604bxll_lqfp100 mpc5604bxlq_lqfp144 mpc5604bxmg_mapbga208 mpc5604cxlh_lqfp64 mpc5604cxll_lqfp100 mpc5605bxll_lqfp100 mpc5605bxlq_lqfp144 mpc5605bxlu_lqfp176 mpc5606bkll_lqfp100
-------------------------	--

Table 2-1. MPC560XB Derivatives

	mpc5606bxlq_lqfp144 mpc5606bxlu_lqfp176 mpc5607bxlu_lqfp176 mpc5607bxmg_mapbga208
--	--

All of the above microcontroller devices are collectively named as MPC560XB .

2.2 Overview

AUTOSAR (AUTomotive Open System ARchitecture) is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

2.3 About this Manual

This Technical Reference employs the following typographical conventions:

Boldface type: Bold is used for important terms, notes and warnings.

Italic font: Italic typeface is used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

2.4 Acronyms and Definitions

Table 2-2. Acronyms and Definitions

Term	Definition
ADC	Analog to Digital Converter
API	Application Programming Interface
ASM	Assembler
AUTOSAR	Automotive Open System Architecture
BSMI	Basic Software Make file Interface
CAN	Controller Area Network
C/CPP	C and C++ Source Code
CS	Chip Select
CTU	Cross Trigger Unit
DEM	Diagnostic Event Manager
DET	Development Error Tracer
DMA	Direct Memory Access
ECU	Electronic Control Unit
FIFO	First In First Out
LSB	Least Significant Bit
MCU	Micro Controller Unit
MIDE	Multi Integrated Development Environment
MSB	Most Significant Bit
N/A	Not Applicable
RAM	Random Access Memory
SIU	Systems Integration Unit
SIUL	Systems Integration Unit Lite
SWS	Software Specification
VLE	Variable Length Encoding
XML	Extensible Markup Language

2.5 Reference List

Table 2-3. Reference List

#	Title	Version
1	AUTOSAR 4.0 Rev0003Port Driver Software Specification Document.	4.0 Rev0003
2	MPC5607B Reference Manual	Rev. 7.2, May 2012
3	MPC5604B Reference Manual	Rev. 8.2, September 2013

Table continues on the next page...

Table 2-3. Reference List (continued)

#	Title	Version
4	5606B Reference Manual	Rev. 2, May 2014
5	5602D Reference Manual	Rev. 4.2, Dec 2013
6	Bolero 1.5M cut 2.2 Errata Sheet Reference Manual	MPC5607B_1M03Y - Rev. 12Dec2013
7	Bolero 512k cut 2.3 Errata Sheet Reference Manual	MPC5604B_2M27V - Rev. 26 FEB 2014
8	Bolero 256k cut 1.1 Errata Sheet Reference Manual	MPC5602D_1M18Y - Rev. , 12 DEC 2013

Chapter 3 Driver

3.1 Requirements

Requirements for this driver are detailed in the AUTOSAR 4.0 Rev0003Port Driver Software Specification document (See Table [Reference List](#)).

3.2 Driver Design Summary

This module provides the service for initializing the whole PORT structure of the microcontroller. Many ports and port pins can be assigned to various functionalities, e.g.

- General purpose I/O
- ADC
- SPI
- SCI
- PWM
- CAN
- LIN
- etc

For this reason, there is an overall configuration and initialization of this port structure. The configuration and mode of these port pins is microcontroller and ECU dependent.

Port initialisation data are written to each port as efficiently as possible. This PORT driver module completes the overall configuration and initialisation of the port structure which is used in the DIO driver module. Therefore, the DIO driver works on pins and ports which are configured by the PORT driver.

The PORT driver is initialised prior to use of the DIO functions. Otherwise DIO functions will exhibit undefined behaviour.

3.3 Deviation from Requirements

The driver deviates from the AUTOSAR Port Driver software specification in some places. Table identifies the AUTOSAR requirements that are not fully implemented, implemented differently, or out of scope for the Port driver. Table [Table 3-1](#) provides Status column description.

Table 3-1. Deviations Status Column Description

Term	Definition
N/A	Not available
N/T	Not testable
N/S	Out of scope
N/I	Not implemented
N/F	Not fully implemented

Below table identifies the AUTOSAR requirements that are not fully implemented, implemented differently, or out of scope for the driver.

Table 3-2. Driver Deviations Table

Requirement	Status	Description	Notes
PORT037	N/I	Production errors shall be reported to Diagnostic Event Manager.	Production errors are not available in Port
PORT041	N/F	Port_Init - This initialization function shall be called to initialize ALL ports and port pins with the configuration set pointed to by ConfigPtr.	User application requirement: User has to configure all available pins
PORT071	N/S	Port_Init: This function shall also be called after a reset, in order to reconfigure the ports and port pins of the MCU.	Application Code Requirement
PORT078	N/S	This function should be called first in order to initialize the port for use. If not called first then no operation can occur on the MCU ports and port pins.	Application Code Requirement
PORT082	N/S	The PORT driver shall not provide for the configuration of Level inversion. The default value shall be set (not inverted). The IO Hardware Abstraction layer shall carry out level Inversion.	Other module Requirement
PORT115	N/A	Values for production code Event Ids are assigned externally by the configuration of the DEM. They are published in the file Dem_IntErrId.h and included via Dem.h.	Production errors are not available in Port

3.4 Software specification

The following sections contains driver software specifications.

3.4.1 Define Reference

Constants supported by the Port driver.

3.4.1.1 Define PORT_E_DIRECTION_UNCHANGEABLE

Port Pin Direction not configured as changeable.

Details:

Det Error value, returned by Port_SetPinDirection if the passed PortPin have unchangeable direction.

Table 3-3. Define PORT_E_DIRECTION_UNCHANGEABLE Description

Name	PORT_E_DIRECTION_UNCHANGEABLE
Initializer	(uint8)0x0B

3.4.1.2 Define PORT_E_INVALID_DIRECTION

API service called with invalid direction.

Details:

Errors and exceptions that will be detected by the PORT driver.

Table 3-4. Define PORT_E_INVALID_DIRECTION Description

Name	PORT_E_INVALID_DIRECTION
Initializer	(uint8)0x11

3.4.1.3 Define PORT_E_MODE_UNCHANGEABLE

API Port_SetPinMode() service called when mode is unchangeable.

Details:

Det Error value, returned by Port_SetPinMode function if the passed PortPin have a unchangeable Mode.

Table 3-5. Define PORT_E_MODE_UNCHANGEABLE Description

Name	PORT_E_MODE_UNCHANGEABLE
Initializer	(uint8)0x0E

3.4.1.4 Define PORT_E_PARAM_CONFIG

API Port_Init() service called with wrong parameter.

Details:

Det Error value, returned by Port_Init function if Port_Init is called with wrong parameter.

Table 3-6. Define PORT_E_PARAM_CONFIG Description

Name	PORT_E_PARAM_CONFIG
Initializer	(uint8)0x0C

3.4.1.5 Define PORT_E_PARAM_INVALID_MODE

API Port_SetPinMode() service called when mode is invalid.

Details:

Det Error value, returned by Port_SetPinMode function if the passed PortPinMode is invalid.

Table 3-7. Define PORT_E_PARAM_INVALID_MODE Description

Name	PORT_E_PARAM_INVALID_MODE
Initializer	(uint8)0x0D

3.4.1.6 Define PORT_E_PARAM_PIN

Invalid Port Pin ID requested.

Details:

Det Error value, returned by Port_SetPinDirection and Port_PinMode if an wrong PortPin ID is passed.

Table 3-8. Define PORT_E_PARAM_PIN Description

Name	PORT_E_PARAM_PIN
Initializer	(uint8)0x0A

3.4.1.7 Define PORT_E_PARAM_POINTER

API service called with NULL Pointer Parameter.

Details:

Det Error value, returned by Port_GetVersionInfo function if API is called with NULL Pointer Parameter.

Table 3-9. Define PORT_E_PARAM_POINTER Description

Name	PORT_E_PARAM_POINTER
Initializer	(uint8)0x10

3.4.1.8 Define PORT_E_UNINIT

API service called without module initialization.

Details:

Det Error value, returned by a function if API service called prior to module initialization.

Table 3-10. Define PORT_E_UNINIT Description

Name	PORT_E_UNINIT
Initializer	(uint8)0x0F

3.4.1.9 Define PORT_GETVERSIONINFO_ID

API service ID for PORT get version info function.

Details:

Parameters used when raising an error/exception.

Table 3-11. Define PORT_GETVERSIONINFO_ID Description

Name	PORT_GETVERSIONINFO_ID
Initializer	(uint8)0x03

3.4.1.10 Define PORT_INIT_ID

API service ID for PORT Init function.

Details:

Parameters used when raising an error/exception.

Table 3-12. Define PORT_INIT_ID Description

Name	PORT_INIT_ID
Initializer	(uint8)0x00

3.4.1.11 Define PORT_REFRESHPINDIRECTION_ID

API service ID for PORT refresh pin direction function.

Details:

Parameters used when raising an error/exception.

Table 3-13. Define PORT_REFRESHPINDIRECTION_ID Description

Name	PORT_REFRESHPINDIRECTION_ID
Initializer	(uint8)0x02

3.4.1.12 Define PORT_SETPINDIRECTION_ID

API service ID for PORT set pin direction function.

Details:

Parameters used when raising an error/exception.

Table 3-14. Define PORT_SETPINDIRECTION_ID Description

Name	PORT_SETPINDIRECTION_ID
Initializer	(uint8)0x01

3.4.1.13 Define PORT_SETPINMODE_ID

API service ID for PORT set pin mode.

Details:

Parameters used when raising an error/exception.

Table 3-15. Define PORT_SETPINMODE_ID Description

Name	PORT_SETPINMODE_ID
Initializer	(uint8)0x04

3.4.2 Macros Reference

Macros supported by the Port driver.

3.4.2.1 Macro SIUL_GPDI

Prototype: SIUL_GPDI (PIN)

3.4.2.2 Macro SIUL_GPDO

Prototype: SIUL_GPDO (PIN)

3.4.2.3 Macro SIUL_IFMC

Prototype: SIUL_IFMC (IFMC_OFFSET)

3.4.2.4 Macro SIUL_MPGPDO

Prototype: SIUL_MPGPDO (MPGPDO_OFFSET)

3.4.2.5 Macro SIUL_PCR

Prototype: SIUL_PCR (PIN)

3.4.2.6 Macro SIUL_PGPDI

Prototype: SIUL_PGPDI (PGPDI_OFFSET)

3.4.2.7 Macro SIUL_PGPDO

Prototype: SIUL_PGPDO (PGPDO_OFFSET)

3.4.2.8 Macro SIUL_PSMI

Prototype: SIUL_PSMI (PSMI_OFFSET)

3.4.3 Enum Reference

Enumeration of all constants supported by the Port driver.

3.4.4 Function Reference

Functions supported by the Port driver.

3.4.4.1 Function FUNC

Sets the port pin direction.

Details:

Returns the version information of this module.

Sets the port pin mode.

Refreshes port direction.

The function `Port_SetPinDirection()` will set the port pin direction during runtime. This function will refresh the direction of all configured ports to the configured direction. The PORT driver will exclude from refreshing those port pins that are configured as "pin direction changeable during runtime".

Pre: `Port_Init()` must have been called first. In order to change the pin direction the `PortPinDirectionChangeable` flag must have been set to `TRUE`.

Implements: DPORT03007

Pre: `Port_Init()` must have been called first.

Implements: DPORT03008

The function `Port_SetPinMode()` will set the port pin mode of the referenced pin during runtime. The function `Port_GetVersionInfo()` will return the version information of this module. The version information includes:

Pre: `Port_Init()` must have been called first.

Implements: DPORT03009

Software specification

- Module Id,
- Vendor Id,
- Vendor specific version numbers.

Implements: DPORT03010

Prototype: `FUNC(void, PORT_CODE);`

Table 3-16. FUNC Arguments

Type	Name	Direction	Description
Pin	input	Pin ID number.	
Direction	input	Port Pin direction.	
Pin	input	Pin ID number.	
Mode	input	New Port Pin mode to be set on port pin.	
pVersionInfo	input, output	Pointer to where to store the version information of this module.	

3.4.5 Structs Reference

Data structures supported by the Port driver.

3.4.5.1 Structure Port_LLD_PinConfigType

Single pin configuration.

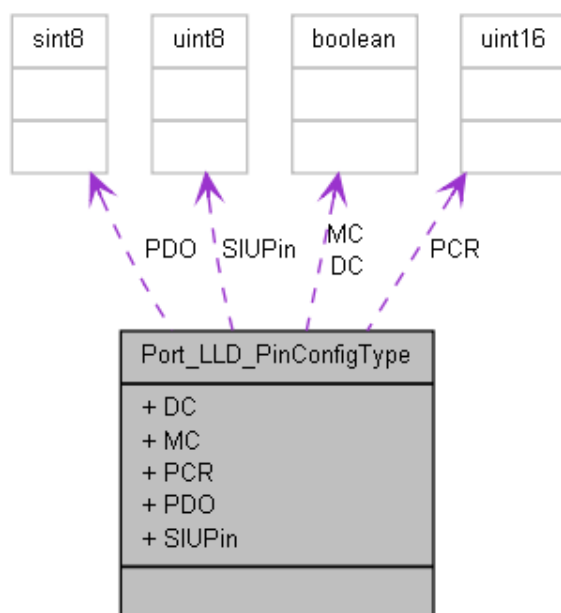


Figure 3-1. Struct Port_LLD_PinConfigType

Details:

This structure contains all configuration parameters of a single pin identified bySIUPin.

Implements: DPORT04028

Declaration:

```

typedef struct
{
    boolean DC,
    boolean MC,
    uint16 PCR,
    sint8 PDO,
    uint8 SIUPin
} Port_LLD_PinConfigType;

```

Table 3-17. Structure Port_LLD_PinConfigType member description

Member	Description
DC	Direction changeability.
MC	Mode changeability.
PCR	Pad Control Register.
PDO	Pad Data Output.
SIUPin	Pin Defined on Part SIUx.

3.4.5.2 Structure Port_LLD_UnUsedPinConfigType

Default pin configuration.

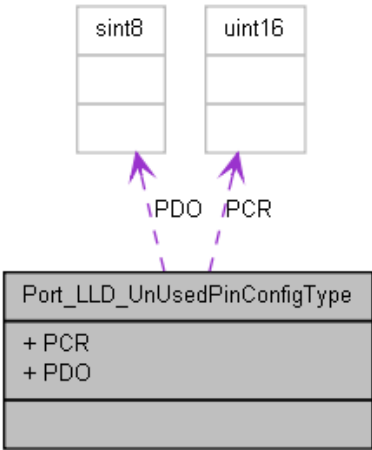


Figure 3-2. Struct Port_LLD_UnUsedPinConfigType

Details:

This structure contains all configuration parameters of a Default pin

Declaration:

```
typedef struct
{
    uint16 PCR,
    sint8 PDO
} Port_LLD_UnUsedPinConfigType;
```

Table 3-18. Structure Port_LLD_UnUsedPinConfigType member description

Member	Description
PCR	
PDO	

3.4.6 Types Reference

Types structures supported by the Port driver.

3.4.7 Variables Reference

Variables supported by the Port driver.

3.4.8 Port_SetPinMode function parameters

This section contains information about how to use the Port_SetPinMode function. In particular, the table below shows all the possible values that the user can use by calling the Port_SetPinMode function.

Pin: This value is the PortPin Id defined in the configuration.

Mode: Is the Mode that the user wishes to change for the selected Pin.

The table is indexed by PortPin Pcr which is associated to a PortPin Id in the configuration.

The screenshot shows a configuration window for a PortPin. The 'General' tab is selected. The configuration parameters are as follows:

Parameter	Value
PortPin Wpe	☐
PortPin Wps	☐
PortPin Ode	☐
PortPin Safe Mode	☐
PortPin Direction Changeable	☐
PortPin Readback	☐
PortPin Mode Changeable	☒
PortPin Id (1 -> 133)	2
PortPinPcr (0 -> 132)	1
PortPin Direction	PORT_PIN_IN
PortPin Initial Mode	PORT_GPIO_MODE
PortPin Mode	GPIO
PortPin Level Value	PORT_PIN_LEVEL_LOW
PortPin Output Slew Rate	SlowestConfiguration

Figure 3-3. PortPin Id and PortPin Pcr

Table 3-19. Mode List

PortPin Pcr	PortPin Mode	Bolero 256k Mode field	Bolero 512k Mode field	Bolero 1.5M Mode field
0	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A0	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	CLKOUT	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	WKUP19	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	eMIOS_A13	-	-	PORT_ALT3_FUNC_MODE
	eMIOS_E0UC13	PORT_ALT3_FUNC_MODE	-	-
	eMIOS_E0UC0	PORT_ALT3_FUNC_MODE	-	-
1	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A1	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	WKUP2	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	NMI_A1	-	-	PORT_ALT2_FUNC_MODE
	eMIOS_E0UC1	PORT_ALT1_FUNC_MODE	-	-
2	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A2	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	WKUP3	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	ADC0_MA2	PORT_ALT3_FUNC_MODE	-	PORT_ALT3_FUNC_MODE
	eMIOS_E0UC2	PORT_ALT1_FUNC_MODE	-	-
3	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A3	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	EIRQ0	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE
	LIN5_TXD	-	-	PORT_ALT2_FUNC_MODE
	SPIx_CS4	PORT_ALT3_FUNC_MODE	-	PORT_ALT3_FUNC_MODE
	ADC1_ANS0	PORT_ANALOG_INPUT_MODE	-	PORT_ANALOG_INPUT_MODE
	eMIOS_E0UC3	PORT_ALT1_FUNC_MODE	-	-
4	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A4	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	WKUP9	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	SPI1_CS0	PORT_ALT3_FUNC_MODE	-	PORT_ALT3_FUNC_MODE
	LIN5_RXD	-	-	PORT_ONLY_INPUT_MODE
	eMIOS_E0UC4	PORT_ALT1_FUNC_MODE	-	-
5	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A5	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	LIN4_TXD	-	-	PORT_ALT2_FUNC_MODE
	eMIOS_E0UC5	PORT_ALT1_FUNC_MODE	-	-

Table continues on the next page...

Table 3-19. Mode List (continued)

PortPin Pcr	PortPin Mode	Bolero 256k Mode field	Bolero 512k Mode field	Bolero 1.5M Mode field
6	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A6	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	EIRQ1	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE
	SPI1_CS1	PORT_ALT3_FUNC_MODE	-	PORT_ALT3_FUNC_MODE
	LIN4_RXD	-	-	PORT_ONLY_INPUT_MODE
	eMIOS_E0UC6	PORT_ALT1_FUNC_MODE	-	-
7	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A7	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	EIRQ2	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE
	LIN3_TXD	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	ADC1_ANS1	PORT_ANALOG_INPUT_MODE	-	PORT_ANALOG_INPUT_MODE
	eMIOS_E0UC7	PORT_ALT1_FUNC_MODE	-	-
8	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A8	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	EIRQ3	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE
	eMIOS_A14	-	-	PORT_ALT2_FUNC_MODE
	LIN3_RXD	-	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	eMIOS_E0UC8	PORT_ALT1_FUNC_MODE	-	-
	eMIOS_E0UC14	PORT_ALT2_FUNC_MODE	-	-
9	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A9	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI1_CS2	PORT_ALT3_FUNC_MODE	-	PORT_ALT3_FUNC_MODE
	eMIOS_E0UC9	PORT_ALT1_FUNC_MODE	-	-
10	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A10	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	I2C0_SDA	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	LIN2_TXD	-	-	PORT_ALT3_FUNC_MODE
	ADC1_ANS2	PORT_ANALOG_INPUT_MODE	-	PORT_ANALOG_INPUT_MODE
	eMIOS_E0UC10	PORT_ALT1_FUNC_MODE	-	-
	LINFlex_2_TX	PORT_ALT3_FUNC_MODE	-	-
11	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A11	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	I2C0_SCL	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE

Table continues on the next page...

Table 3-19. Mode List (continued)

PortPin Pcr	PortPin Mode	Bolero 256k Mode field	Bolero 512k Mode field	Bolero 1.5M Mode field
	ADC1_ANS3	PORT_ANALOG_INPUT_MODE	-	PORT_ANALOG_INPUT_MODE
	LIN2_RXD	PORT_ONLY_INPUT_MODE	-	PORT_ONLY_INPUT_MODE
	EIRQ16	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE
	eMIOS_E0UC11	PORT_ALT1_FUNC_MODE	-	-
12	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A28	-	-	PORT_ALT2_FUNC_MODE
	SPI1_CS3	-	-	PORT_ALT3_FUNC_MODE
	EIRQ17	PORT_EXTRA2_INPUT_MODE	-	PORT_EXTRA2_INPUT_MODE
	SPI0_SIN	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
13	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	SPI0_SOUT	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	eMIOS_A29	-	-	PORT_ALT2_FUNC_MODE
	SPI1_CS3	PORT_ALT3_FUNC_MODE	-	-
14	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	SPI0_SCK	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI0_CS0	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	EIRQ4	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE
	eMIOS_A0	-	-	PORT_ALT3_FUNC_MODE
	eMIOS_E0UC0	PORT_ALT3_FUNC_MODE	-	-
15	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	SPI0_CS0	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI0_SCK	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	WKUP10	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	eMIOS_A1	-	-	PORT_ALT3_FUNC_MODE
	eMIOS_E0UC1	PORT_ALT3_FUNC_MODE	-	-
16	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	CAN0_TX	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	eMIOS_A30	-	-	PORT_ALT2_FUNC_MODE
	LIN0_TXD	-	-	PORT_ALT3_FUNC_MODE
	LINFlex_2_TX	PORT_ALT3_FUNC_MODE	-	-
17	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	WKUP4	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	eMIOS_A31	-	-	PORT_ALT2_FUNC_MODE

Table continues on the next page...

Table 3-19. Mode List (continued)

PortPin Pcr	PortPin Mode	Bolero 256k Mode field	Bolero 512k Mode field	Bolero 1.5M Mode field
	LIN0_RXD	-	-	PORT_EXTRA_INPUT_MODE
	CAN0_RX	-	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	LINFlex_0_RX	PORT_ALT3_FUNC_MODE	-	-
	FlexCAN_0_RX	PORT_EXTRA_INPUT_MODE	-	-
18	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	LIN0_TXD	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	I2C0_SDA	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	eMIOS_A30	-	-	PORT_ALT3_FUNC_MODE
19	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	WKUP11	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	eMIOS_A31	-	-	PORT_ALT1_FUNC_MODE
	I2C0_SCL	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	LIN0_RXD	-	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	LINFlex_0_RX	PORT_EXTRA_INPUT_MODE	-	-
20	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	ADCx_ANP0	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
21	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	ADCx_ANP1	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
22	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	ADCx_ANP2	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
23	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	ADCx_ANP3	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
24	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	ADC0_ANS0	-	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
	ADC1_ANS4	PORT_ANALOG_INPUT_MODE	-	PORT_ANALOG_INPUT_MODE
	WKUP25	PORT_ONLY_INPUT_MODE	-	PORT_ONLY_INPUT_MODE
	OSC32K_XTAL	-	PORT_OSC_SIGNAL_MODE	PORT_OSC_SIGNAL_MODE
25	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE

Table continues on the next page...

Table 3-19. Mode List (continued)

PortPin Pcr	PortPin Mode	Bolero 256k Mode field	Bolero 512k Mode field	Bolero 1.5M Mode field
	ADC0_ANS1	-	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
	ADC1_ANS5	PORT_ANALOG_INPUT_MODE	-	PORT_ANALOG_INPUT_MODE
	WKUP26	PORT_ONLY_INPUT_MODE	-	PORT_ONLY_INPUT_MODE
	OSC32K_EXTAL	-	PORT_OSC_SIGNAL_MODE	PORT_OSC_SIGNAL_MODE
26	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	ADC0_ANS2	-	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
	ADC1_ANS6	PORT_ANALOG_INPUT_MODE	-	PORT_ANALOG_INPUT_MODE
	WKUP8	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
27	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A3	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI0_CS0	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE
	ADC0_ANS3	-	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
	eMIOS_E0UC3	PORT_ALT3_FUNC_MODE	-	-
	ADC1_S12	PORT_ANALOG_INPUT_MODE	-	-
28	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A4	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI0_CS1	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE
	ADC0_ANX0	-	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
	eMIOS_E0UC4	PORT_ALT1_FUNC_MODE	-	-
	ADC1_X0	PORT_ANALOG_INPUT_MODE	-	-
29	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A5	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI0_CS2	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE
	ADC0_ANX1	-	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
	eMIOS_E0UC5	PORT_ALT1_FUNC_MODE	-	-
	ADC1_X1	PORT_ANALOG_INPUT_MODE	-	-
30	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A6	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI0_CS3	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE

Table continues on the next page...

Table 3-19. Mode List (continued)

PortPin Pcr	PortPin Mode	Bolero 256k Mode field	Bolero 512k Mode field	Bolero 1.5M Mode field
	ADC0_ANX2	-	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
	eMIOS_E0UC6	PORT_ALT1_FUNC_MODE	-	-
	ADC1_X2	PORT_ANALOG_INPUT_MODE	-	-
31	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A7	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI0_CS4	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE
	ADC0_ANX3	-	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
	eMIOS_E0UC7	PORT_ALT1_FUNC_MODE	-	-
	ADC1_X3	PORT_ANALOG_INPUT_MODE	-	-
32	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	JTAG_TDI	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
33	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	JTAG_TDO	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
34	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	SPI1_SCK	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	CAN4_TX	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	EIRQ5	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE
	DEBUG0	-	-	PORT_ALT3_FUNC_MODE
35	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	SPI1_CS0	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	ADC0_MA0	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	EIRQ6	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE
	DEBUG1	-	-	PORT_ALT3_FUNC_MODE
	CAN1_RX	-	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	CAN4_RX	-	PORT_EXTRA_INPUT_MODE	PORT_EXTRA_INPUT_MODE
36	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_B31	-	-	PORT_ALT1_FUNC_MODE
	DEBUG2	-	-	PORT_ALT3_FUNC_MODE
	EIRQ18	PORT_EXTRA2_INPUT_MODE	-	PORT_EXTRA2_INPUT_MODE
	SPI1_SIN	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	CAN3_RX	-	PORT_EXTRA_INPUT_MODE	PORT_EXTRA_INPUT_MODE

Table continues on the next page...

Table 3-19. Mode List (continued)

PortPin Pcr	PortPin Mode	Bolero 256k Mode field	Bolero 512k Mode field	Bolero 1.5M Mode field
37	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	SPI1_SOUT	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	CAN3_TX	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	EIRQ7	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE
	DEBUG3	-	-	PORT_ALT3_FUNC_MODE
38	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	LIN1_TXD	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	eMIOS_B28	-	-	PORT_ALT2_FUNC_MODE
	DEBUG4	-	-	PORT_ALT3_FUNC_MODE
39	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	WKUP12	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	eMIOS_B29	-	-	PORT_ALT2_FUNC_MODE
	LIN1_RXD	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	DEBUG5	-	-	PORT_ALT3_FUNC_MODE
40	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	LIN2_TXD	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	eMIOS_A3	-	-	PORT_ALT2_FUNC_MODE
	eMIOS_E0UC3	PORT_ALT2_FUNC_MODE	-	-
	DEBUG6	-	-	PORT_ALT3_FUNC_MODE
41	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	WKUP13	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	eMIOS_A7	-	-	PORT_ALT2_FUNC_MODE
	eMIOS_E0UC7	PORT_ALT2_FUNC_MODE	-	-
	DEBUG7	-	-	PORT_ALT3_FUNC_MODE
	LIN2_RXD	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
42	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	CAN1_TX	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	CAN4_TX	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	ADC0_MA1	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE
43	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	ADC0_MA2	PORT_ALT3_FUNC_MODE	-	PORT_ALT3_FUNC_MODE
	CAN1_RX	-	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	CAN4_RX	-	PORT_EXTRA_INPUT_MODE	PORT_EXTRA_INPUT_MODE

Table continues on the next page...

Table 3-19. Mode List (continued)

PortPin Pcr	PortPin Mode	Bolero 256k Mode field	Bolero 512k Mode field	Bolero 1.5M Mode field
	WKUP5	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
44	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A12	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI2_SIN	-	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	eMIOS_E0UC12	PORT_ALT1_FUNC_MODE	-	-
	EIRQ19	PORT_EXTRA2_INPUT_MODE	-	PORT_EXTRA2_INPUT_MODE
45	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A13	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI2_SOUT	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	eMIOS_E0UC13	PORT_ALT1_FUNC_MODE	-	-
46	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A14	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI2_SCK	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	eMIOS_E0UC14	PORT_ALT1_FUNC_MODE	-	-
	EIRQ8	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE
47	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A15	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI2_CS0	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	eMIOS_E0UC15	PORT_ALT1_FUNC_MODE	-	-
	EIRQ20	PORT_EXTRA2_INPUT_MODE	-	PORT_EXTRA2_INPUT_MODE
48	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	ADCx_ANP4	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
	WKUP27	PORT_ONLY_INPUT_MODE	-	PORT_ONLY_INPUT_MODE
49	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	ADCx_ANP5	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
	WKUP28	PORT_ONLY_INPUT_MODE	-	PORT_ONLY_INPUT_MODE
50	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	ADCx_ANP6	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
51	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	ADCx_ANP7	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
52	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE

Table continues on the next page...

Table 3-19. Mode List (continued)

PortPin Pcr	PortPin Mode	Bolero 256k Mode field	Bolero 512k Mode field	Bolero 1.5M Mode field
	ADCx_ANP8	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
53	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	ADCx_ANP9	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
54	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	ADCx_ANP10	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
55	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	ADCx_ANP11	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
56	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	ADCx_ANP12	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
57	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	ADCx_ANP13	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
58	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	ADCx_ANP14	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
59	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	ADCx_ANP15	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
60	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	SPI0_CS5	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	eMIOS_A24	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	ADC0_ANS4	-	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
	eMIOS_E0UC24	PORT_ALT2_FUNC_MODE	-	-
	ADC1_S8	PORT_ANALOG_INPUT_MODE	-	-
61	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	SPI1_CS0	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	eMIOS_A25	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	ADC0_ANS5	-	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
	eMIOS_E0UC25	PORT_ALT2_FUNC_MODE	-	-
	ADC1_S9	PORT_ANALOG_INPUT_MODE	-	-
62	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	SPI1_CS1	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	eMIOS_A26	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE

Table continues on the next page...

Table 3-19. Mode List (continued)

PortPin Pcr	PortPin Mode	Bolero 256k Mode field	Bolero 512k Mode field	Bolero 1.5M Mode field
	ADC0_ANS6	-	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
	eMIOS_E0UC26	PORT_ALT2_FUNC_MODE	-	-
	ADC1_S10	PORT_ANALOG_INPUT_MODE	-	-
63	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	SPI1_CS2	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	eMIOS_A27	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	ADC0_ANS7	-	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
	eMIOS_E0UC27	PORT_ALT2_FUNC_MODE	-	-
	ADC1_S11	PORT_ANALOG_INPUT_MODE	-	-
64	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A16	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	CAN5_RX	-	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	eMIOS_E0UC16	PORT_ALT1_FUNC_MODE	-	-
	WKUP6	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
65	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A17	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	CAN5_TX	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	eMIOS_E0UC17	PORT_ALT1_FUNC_MODE	-	-
66	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A18	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI1_SIN	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	eMIOS_E0UC18	PORT_ALT1_FUNC_MODE	-	-
	EIRQ21	PORT_EXTRA2_INPUT_MODE	-	PORT_EXTRA2_INPUT_MODE
67	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A19	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI1_SOUT	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	eMIOS_E0UC19	PORT_ALT1_FUNC_MODE	-	-
68	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A20	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI1_SCK	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	eMIOS_E0UC20	PORT_ALT1_FUNC_MODE	-	-
	EIRQ9	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE

Table continues on the next page...

Table 3-19. Mode List (continued)

PortPin Pcr	PortPin Mode	Bolero 256k Mode field	Bolero 512k Mode field	Bolero 1.5M Mode field
69	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A21	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI1_CS0	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	ADC0_MA2	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE
	eMIOS_E0UC21	PORT_ALT1_FUNC_MODE	-	-
70	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A22	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI0_CS3	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	ADC0_MA1	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE
	EIRQ22	PORT_EXTRA2_INPUT_MODE	-	PORT_EXTRA2_INPUT_MODE
	eMIOS_E0UC22	PORT_ALT1_FUNC_MODE	-	-
71	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A23	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI0_CS2	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	ADC0_MA0	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE
	EIRQ23	PORT_EXTRA2_INPUT_MODE	-	PORT_EXTRA2_INPUT_MODE
	eMIOS_E0UC23	PORT_ALT1_FUNC_MODE	-	-
72	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	CAN2_TX	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	eMIOS_A22	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	CAN3_TX	-	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE
	eMIOS_E0UC22	PORT_ALT2_FUNC_MODE	-	-
73	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	CAN2_RX	-	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	eMIOS_A23	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	CAN3_RX	-	PORT_EXTRA_INPUT_MODE	PORT_EXTRA_INPUT_MODE
	WKUP7	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	eMIOS_E0UC23	PORT_ALT2_FUNC_MODE	-	-
74	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	LIN3_TXD	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI1_CS3	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	EIRQ10	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE
	eMIOS_B30	-	-	PORT_ALT3_FUNC_MODE
75	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE

Table continues on the next page...

Table 3-19. Mode List (continued)

PortPin Pcr	PortPin Mode	Bolero 256k Mode field	Bolero 512k Mode field	Bolero 1.5M Mode field
	eMIOS_A24	-	-	PORT_ALT1_FUNC_MODE
	SPI1_CS4	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	LIN3_RXD	-	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	WKUP14	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	eMIOS_E0UC24	PORT_ALT1_FUNC_MODE	-	-
76	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_B19	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	ADC1_ANS7	PORT_ANALOG_INPUT_MODE	-	PORT_ANALOG_INPUT_MODE
	EIRQ11	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE
	SPI2_SIN	-	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
77	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	SPI2_SOUT	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	eMIOS_B20	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
78	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	SPI2_SCK	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	eMIOS_B21	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	EIRQ12	-	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE
79	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	SPI2_CS0	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	eMIOS_B22	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
80	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A10	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI1_CS3	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	ADC0_ANS8	-	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
81	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A11	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI1_CS4	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	ADC0_ANS9	-	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
82	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A12	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI2_CS0	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	ADC0_ANS10	-	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE

Table continues on the next page...

Table 3-19. Mode List (continued)

PortPin Pcr	PortPin Mode	Bolero 256k Mode field	Bolero 512k Mode field	Bolero 1.5M Mode field
83	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A13	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI2_CS1	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	ADC0_ANS11	-	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
84	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A14	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI2_CS2	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	ADC0_ANS12	-	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
85	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A22	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI2_CS3	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	ADC0_ANS13	-	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
86	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A23	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI1_CS1	-	-	PORT_ALT2_FUNC_MODE
	ADC0_ANS14	-	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
87	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	SPI1_CS2	-	-	PORT_ALT2_FUNC_MODE
	ADC0_ANS15	-	PORT_ANALOG_INPUT_MODE	PORT_ANALOG_INPUT_MODE
88	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	CAN3_TX	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI0_CS4	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	CAN2_TX	-	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE
89	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_B1	-	-	PORT_ALT1_FUNC_MODE
	SPI0_CS5	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	CAN2_RX	-	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	CAN3_RX	-	PORT_EXTRA_INPUT_MODE	PORT_EXTRA_INPUT_MODE
90	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	SPI0_CS1	-	-	PORT_ALT1_FUNC_MODE
	LIN4_TXD	-	-	PORT_ALT2_FUNC_MODE
	eMIOS_B2	-	-	PORT_ALT3_FUNC_MODE
91	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE

Table continues on the next page...

Table 3-19. Mode List (continued)

PortPin Pcr	PortPin Mode	Bolero 256k Mode field	Bolero 512k Mode field	Bolero 1.5M Mode field
	WKUP15	-	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	SPI0_CS2	-	-	PORT_ALT1_FUNC_MODE
	eMIOS_B3	-	-	PORT_ALT2_FUNC_MODE
	LIN4_RXD	-	-	PORT_ONLY_INPUT_MODE
92	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_B25	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	LIN5_TXD	-	-	PORT_ALT2_FUNC_MODE
93	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	WKUP16	-	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	eMIOS_B26	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	LIN5_RXD	-	-	PORT_ONLY_INPUT_MODE
94	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	CAN4_TX	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	eMIOS_B27	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	CAN1_TX	-	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE
95	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_B4	-	-	PORT_ALT1_FUNC_MODE
	CAN1_RX	-	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	CAN4_RX	-	PORT_EXTRA_INPUT_MODE	PORT_EXTRA_INPUT_MODE
	EIRQ13	-	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE
96	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	CAN5_TX	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	eMIOS_B23	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
97	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_B24	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	CAN5_RX	-	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	EIRQ14	-	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE
98	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_B11	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI3_SOUT	-	-	PORT_ALT2_FUNC_MODE
99	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_B12	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE

Table continues on the next page...

Table 3-19. Mode List (continued)

PortPin Pcr	PortPin Mode	Bolero 256k Mode field	Bolero 512k Mode field	Bolero 1.5M Mode field
100	WKUP17	-	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
	SPI3_CS0	-	-	PORT_ALT2_FUNC_MODE
	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_B13	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI3_SCK	-	-	PORT_ALT2_FUNC_MODE
101	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_B14	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	WKUP18	-	-	PORT_ONLY_INPUT_MODE
	SPI3_SIN	-	-	PORT_ALT2_FUNC_MODE
102	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_B15	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	LIN6_TXD	-	-	PORT_ALT2_FUNC_MODE
103	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_B16	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	eMIOS_B30	-	-	PORT_ALT2_FUNC_MODE
	WKUP20	-	-	PORT_ONLY_INPUT_MODE
	LIN6_RXD	-	-	PORT_ONLY_INPUT_MODE
104	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_B17	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	LIN7_TXD	-	-	PORT_ALT2_FUNC_MODE
	SPI2_CS0	-	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE
	EIRQ15	-	PORT_EXTRA2_INPUT_MODE	PORT_EXTRA2_INPUT_MODE
105	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_B18	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI2_SCK	-	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE
	WKUP21	-	-	PORT_ONLY_INPUT_MODE
	LIN7_RXD	-	-	PORT_ONLY_INPUT_MODE
106	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A24	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	eMIOS_B31	-	-	PORT_ALT2_FUNC_MODE
	SPI4_SIN	-	-	PORT_ONLY_INPUT_MODE
107	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A25	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE

Table continues on the next page...

Table 3-19. Mode List (continued)

PortPin Pcr	PortPin Mode	Bolero 256k Mode field	Bolero 512k Mode field	Bolero 1.5M Mode field
	SPI4_CS0	-	-	PORT_ALT2_FUNC_MODE
108	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A26	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI4_SOUT	-	-	PORT_ALT2_FUNC_MODE
109	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_A27	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI4_SCK	-	-	PORT_ALT2_FUNC_MODE
110	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_B0	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	LIN8_TXD	-	-	PORT_ALT2_FUNC_MODE
111	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_B1	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	LIN8_RXD	-	-	PORT_ONLY_INPUT_MODE
112	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_B2	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI1_SIN	-	PORT_ONLY_INPUT_MODE	PORT_ONLY_INPUT_MODE
113	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_B3	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI1_SOUT	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
114	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_B4	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI1_SCK	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
115	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_B5	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI1_CS0	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
116	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_B6	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
117	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_B7	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
118	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_B8	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	ADC0_MA2	-	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE
119	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE
	eMIOS_B9	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI2_CS3	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	ADC0_MA1	-	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE
120	GPIO	-	PORT_GPIO_MODE	PORT_GPIO_MODE

Table continues on the next page...

Table 3-19. Mode List (continued)

PortPin Pcr	PortPin Mode	Bolero 256k Mode field	Bolero 512k Mode field	Bolero 1.5M Mode field
	eMIOS_B10	-	PORT_ALT1_FUNC_MODE	PORT_ALT1_FUNC_MODE
	SPI2_CS2	-	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
	ADC0_MA0	-	PORT_ALT3_FUNC_MODE	PORT_ALT3_FUNC_MODE
121	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	JTAG_TCK	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
122	GPIO	PORT_GPIO_MODE	PORT_GPIO_MODE	PORT_GPIO_MODE
	JTAG_TMS	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE	PORT_ALT2_FUNC_MODE
123	GPIO	-	-	PORT_GPIO_MODE
	SPI3_SOUT	-	-	PORT_ALT1_FUNC_MODE
	SPI4_CS0	-	-	PORT_ALT2_FUNC_MODE
	eMIOS_B5	-	-	PORT_ALT3_FUNC_MODE
124	GPIO	-	-	PORT_GPIO_MODE
	SPI3_SCK	-	-	PORT_ALT1_FUNC_MODE
	SPI4_CS1	-	-	PORT_ALT2_FUNC_MODE
	eMIOS_B25	-	-	PORT_ALT3_FUNC_MODE
125	GPIO	-	-	PORT_GPIO_MODE
	SPI4_SOUT	-	-	PORT_ALT1_FUNC_MODE
	SPI3_CS0	-	-	PORT_ALT2_FUNC_MODE
	eMIOS_B26	-	-	PORT_ALT3_FUNC_MODE
126	GPIO	-	-	PORT_GPIO_MODE
	SPI4_SCK	-	-	PORT_ALT1_FUNC_MODE
	SPI3_CS1	-	-	PORT_ALT2_FUNC_MODE
	eMIOS_B27	-	-	PORT_ALT3_FUNC_MODE
127	GPIO	-	-	PORT_GPIO_MODE
	SPI5_SOUT	-	-	PORT_ALT1_FUNC_MODE
	eMIOS_B17	-	-	PORT_ALT3_FUNC_MODE
128	GPIO	-	-	PORT_GPIO_MODE
	eMIOS_A28	-	-	PORT_ALT1_FUNC_MODE
	LIN8_TXD	-	-	PORT_ALT2_FUNC_MODE
129	GPIO	-	-	PORT_GPIO_MODE
	eMIOS_A29	-	-	PORT_ALT1_FUNC_MODE
	LIN8_RXD	-	-	PORT_ONLY_INPUT_MODE
	WKUP24	-	-	PORT_ONLY_INPUT_MODE
130	GPIO	-	-	PORT_GPIO_MODE
	eMIOS_A30	-	-	PORT_ALT1_FUNC_MODE
	LIN9_TXD	-	-	PORT_ALT2_FUNC_MODE
131	GPIO	-	-	PORT_GPIO_MODE

Table continues on the next page...

Table 3-19. Mode List (continued)

PortPin Pcr	PortPin Mode	Bolero 256k Mode field	Bolero 512k Mode field	Bolero 1.5M Mode field
	eMIOS_A31	-	-	PORT_ALT1_FUNC_MODE
	LIN9_RXD	-	-	PORT_ONLY_INPUT_MODE
	WKUP23	-	-	PORT_ONLY_INPUT_MODE
132	GPIO	-	-	PORT_GPIO_MODE
	eMIOS_B28	-	-	PORT_ALT1_FUNC_MODE
	SPI4_SOUT	-	-	PORT_ALT2_FUNC_MODE
133	GPIO	-	-	PORT_GPIO_MODE
	eMIOS_B29	-	-	PORT_ALT1_FUNC_MODE
	SPI4_SCK	-	-	PORT_ALT2_FUNC_MODE
134	GPIO	-	-	PORT_GPIO_MODE
	eMIOS_B30	-	-	PORT_ALT1_FUNC_MODE
	SPI4_CS0	-	-	PORT_ALT2_FUNC_MODE
135	GPIO	-	-	PORT_GPIO_MODE
	eMIOS_B31	-	-	PORT_ALT1_FUNC_MODE
	SPI4_CS1	-	-	PORT_ALT2_FUNC_MODE
136	GPIO	-	-	PORT_GPIO_MODE
	ADC0_ANS16	-	-	PORT_ANALOG_INPUT_MODE
137	GPIO	-	-	PORT_GPIO_MODE
	ADC0_ANS17	-	-	PORT_ANALOG_INPUT_MODE
138	GPIO	-	-	PORT_GPIO_MODE
	ADC0_ANS18	-	-	PORT_ANALOG_INPUT_MODE
139	GPIO	-	-	PORT_GPIO_MODE
	ADC0_ANS19	-	-	PORT_ANALOG_INPUT_MODE
	SPI3_SIN	-	-	PORT_ONLY_INPUT_MODE
140	GPIO	-	-	PORT_GPIO_MODE
	SPI3_CS0	-	-	PORT_ALT1_FUNC_MODE
	ADC0_ANS20	-	-	PORT_ANALOG_INPUT_MODE
141	GPIO	-	-	PORT_GPIO_MODE
	SPI3_CS1	-	-	PORT_ALT1_FUNC_MODE
	ADC0_ANS21	-	-	PORT_ANALOG_INPUT_MODE
142	GPIO	-	-	PORT_GPIO_MODE
	SPI4_SIN	-	-	PORT_ONLY_INPUT_MODE

Table continues on the next page...

Table 3-19. Mode List (continued)

PortPin Pcr	PortPin Mode	Bolero 256k Mode field	Bolero 512k Mode field	Bolero 1.5M Mode field
	ADC0_ANS22	-	-	PORT_ANALOG_INPUT_MODE
143	GPIO	-	-	PORT_GPIO_MODE
	SPI4_CS0	-	-	PORT_ALT1_FUNC_MODE
	ADC0_ANS23	-	-	PORT_ANALOG_INPUT_MODE
144	GPIO	-	-	PORT_GPIO_MODE
	SPI4_CS1	-	-	PORT_ALT1_FUNC_MODE
	ADC0_ANS24	-	-	PORT_ANALOG_INPUT_MODE
145	GPIO	-	-	PORT_GPIO_MODE
	SPI5_SIN	-	-	PORT_ONLY_INPUT_MODE
	ADC0_ANS25	-	-	PORT_ANALOG_INPUT_MODE
146	GPIO	-	-	PORT_GPIO_MODE
	SPI5_CS0	-	-	PORT_ALT1_FUNC_MODE
	ADC0_ANS26	-	-	PORT_ANALOG_INPUT_MODE
147	GPIO	-	-	PORT_GPIO_MODE
	SPI5_CS1	-	-	PORT_ALT1_FUNC_MODE
	ADC0_ANS27	-	-	PORT_ANALOG_INPUT_MODE
148	GPIO	-	-	PORT_GPIO_MODE
	SPI5_SCK	-	-	PORT_ALT1_FUNC_MODE
	eMIOS_B18	-	-	PORT_ALT2_FUNC_MODE

3.5 Symbolic Names Disclaimer

All containers having the symbolic name tag set as true in the Autosar schema will generate defines like:

```
#define <Container_Short_Name> <Container_ID>
```

For this reason it is forbidden to duplicate the name of such containers across the MCAL configuration, or to use names that may trigger other compile issues (e.g. match existing #ifdefs arguments).

Chapter 4

Tresos Configuration Plug-in

This chapter describes the Tresos configuration plug-in for the Port Driver. The most of the parameters are described below.

4.1 Configuration elements of Port

Included forms :

- IMPLEMENTATION_CONFIG_VARIANT
- PortGeneral
- CommonPublishedInformation
- PortConfigSet

Table 4-1. Revision table

Revision	Date
4.1.0	2010-12-03

4.2 Form IMPLEMENTATION_CONFIG_VARIANT

VariantPreCompile: Only precompile time configuration parameters. Only one set of parameters. VariantPostBuild: Mix of precompile and postbuild time configuration parameters. More sets of parameters. If Config Variant = VariantPreCompile, the files Port_Cfg.h and Port_Cfg.c should be used. If Config Variant = VariantPostBuild, the files Port_Cfg.h and Port_PBcfg.c should be used.

Port

Name Port

General PortConfigSet Published Information

Config Variant*

PortGeneral

PortGeneral

Port Development Error Detect

Port SetPinDirection Api

Port SetPinMode Api

Port VersionInfo Api

Figure 4-1. Tresos Plugin snapshot for IMPLEMENTATION_CONFIG_VARIANT form.

Table 4-2. Attribute IMPLEMENTATION_CONFIG_VARIANT detailed description

Property	Value
Label	Config Variant
Type	ENUMERATION
Default	VariantPostBuild
Range	VariantPostBuild VariantPreCompile

4.3 Form PortGeneral

Module wide configuration parameters of the PORT driver.

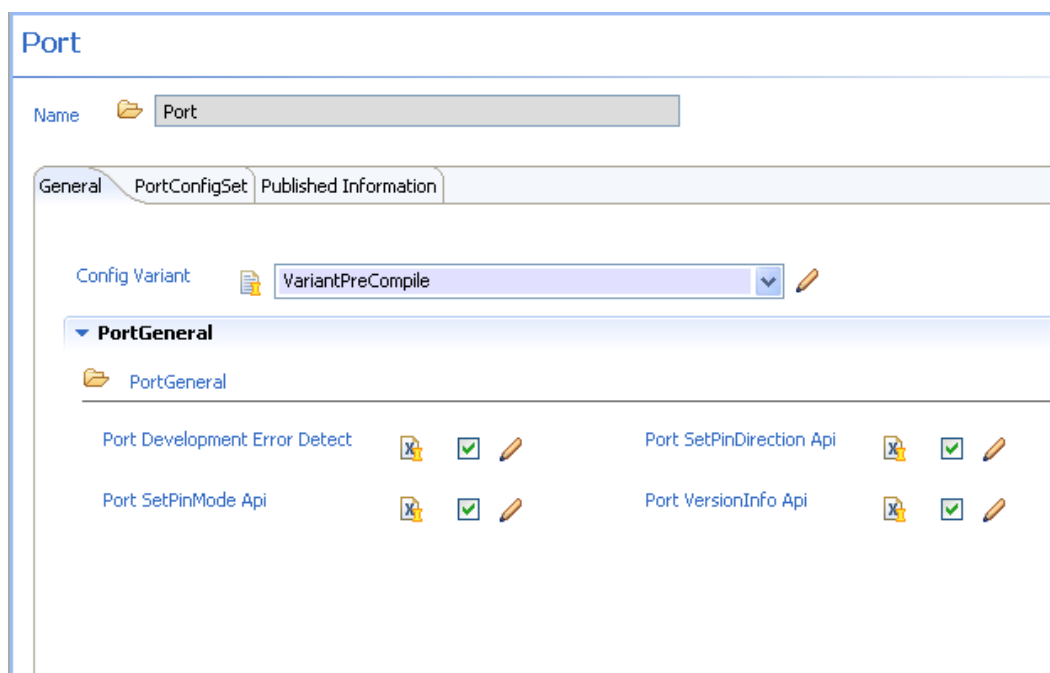


Figure 4-2. Tresos Plugin snapshot for PortGeneral form.

4.3.1 PortDevErrorDetect (PortGeneral)

Switches the Development Error Detection and Notification ON or OFF.

Table 4-3. Attribute PortDevErrorDetect (PortGeneral) detailed description

Property	Value
Label	Port Development Error Detect
Type	BOOLEAN
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	true

4.3.2 PortSetPinDirectionApi (PortGeneral)

Pre-processor switch to enable/disable the use of the function Port_SetPinDirection().

Table 4-4. Attribute PortSetPinDirectionApi (PortGeneral) detailed description

Property	Value
Label	Port SetPinDirection Api

Table continues on the next page...

Table 4-4. Attribute PortSetPinDirectionApi (PortGeneral) detailed description (continued)

Property	Value
Type	BOOLEAN
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	true

4.3.3 PortSetPinModeApi (PortGeneral)

Pre-processor switch to enable/disable the use of the function Port_SetPinMode().

Table 4-5. Attribute PortSetPinModeApi (PortGeneral) detailed description

Property	Value
Label	Port SetPinMode Api
Type	BOOLEAN
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	true

4.3.4 PortVersionInfoApi (PortGeneral)

Pre-processor switch to enable/disable the API to read out the modules version information.

Table 4-6. Attribute PortVersionInfoApi (PortGeneral) detailed description

Property	Value
Label	Port VersionInfo Api
Type	BOOLEAN
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	true

4.3.5 PortPinAbstractedMode (PortGeneral)

Enable/Disable the Abstraction Mode

Table 4-7. Attribute PortPinAbstractedMode (PortGeneral) detailed description

Property	Value
Label	PortPin Abstracted Mode
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.4 Form CommonPublishedInformation

Common container, aggregated by all modules. It contains published information about vendor and versions.

CommonPublishedInformation	
CommonPublishedInformation	
AUTOSAR Major Version	4
AUTOSAR Minor Version	0
AUTOSAR Revision Version	3
Module Id.	124
Software Major Version	0
Software Minor Version	9
Software Patch Version	0
VendorApiInfix	
VendorId	43

Figure 4-3. Tresos Plugin snapshot for CommonPublishedInformation form.

4.4.1 ArReleaseMajorVersion (CommonPublishedInformation)

Major version number of AUTOSAR specification on which the appropriate implementation is based on.

**Table 4-8. Attribute ArReleaseMajorVersion
(CommonPublishedInformation) detailed description**

Property	Value
Label	AUTOSAR Major Version
Type	INTEGER_LABEL
Origin	Custom
Symbolic Name	false
Default	4
Invalid	Range >=4 <=4

4.4.2 ArReleaseMinorVersion (CommonPublishedInformation)

Minor version number of AUTOSAR specification on which the appropriate implementation is based on.

**Table 4-9. Attribute ArReleaseMinorVersion
(CommonPublishedInformation) detailed description**

Property	Value
Label	AUTOSAR Minor Version
Type	INTEGER_LABEL
Origin	Custom
Symbolic Name	false
Default	0
Invalid	Range >=0 <=0

4.4.3 ArReleaseRevisionVersion (CommonPublishedInformation)

Revision version number of AUTOSAR specification on which the appropriate implementation is based on.

**Table 4-10. Attribute ArReleaseRevisionVersion
(CommonPublishedInformation) detailed description**

Property	Value
Label	AUTOSAR Revision Version
Type	INTEGER_LABEL

Table continues on the next page...

Table 4-10. Attribute ArReleaseRevisionVersion (CommonPublishedInformation) detailed description (continued)

Property	Value
Origin	Custom
Symbolic Name	false
Default	3
Invalid	Range >=3 <=3

4.4.4 ModuleId (CommonPublishedInformation)

Module ID of this module from Module List.

Table 4-11. Attribute ModuleId (CommonPublishedInformation) detailed description

Property	Value
Label	Module Id.
Type	INTEGER_LABEL
Origin	Custom
Symbolic Name	false
Default	124
Invalid	Range >=124 <=124

4.4.5 SwMajorVersion (CommonPublishedInformation)

Major version number of the vendor specific implementation of the module. The numbering is vendor specific.

Table 4-12. Attribute SwMajorVersion (CommonPublishedInformation) detailed description

Property	Value
Label	Software Major Version
Type	INTEGER_LABEL
Origin	Custom
Symbolic Name	false
Default	1

Table continues on the next page...

Table 4-12. Attribute SwMajorVersion (CommonPublishedInformation) detailed description (continued)

Property	Value
Invalid	Range ≥ 1 ≤ 1

4.4.6 SwMinorVersion (CommonPublishedInformation)

Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.

Table 4-13. Attribute SwMinorVersion (CommonPublishedInformation) detailed description

Property	Value
Label	Software Minor Version
Type	INTEGER_LABEL
Origin	Custom
Symbolic Name	false
Default	0
Invalid	Range ≥ 0 ≤ 0

4.4.7 SwPatchVersion (CommonPublishedInformation)

Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.

Table 4-14. Attribute SwPatchVersion (CommonPublishedInformation) detailed description

Property	Value
Label	Software Patch Version
Type	INTEGER_LABEL
Origin	Custom
Symbolic Name	false
Default	1
Invalid	Range ≥ 1 ≤ 1

4.4.8 VendorApiInfix (CommonPublishedInformation)

In driver modules which can be instantiated several times on a single ECU, BSW00347 requires that the name of APIs is extended by the VendorId and a vendor specific name. This parameter is used to specify the vendor specific name. In total, the implementation specific name is generated as follows:

<ModuleName>_>VendorId>_<VendorApiInfix><Api name from SWS>. E.g. assuming that the VendorId of the implementor is 123 and the implementer chose a VendorApiInfix of "v11r456" a api name Can_Write defined in the SWS will translate to Can_123_v11r456Write. This parameter is mandatory for all modules with upper multiplicity > 1. It shall not be used for modules with upper multiplicity =1.

Table 4-15. Attribute VendorApiInfix (CommonPublishedInformation) detailed description

Property	Value
Type	STRING_LABEL
Origin	Custom
Symbolic Name	false
Default	
Enable	false

4.4.9 VendorId (CommonPublishedInformation)

Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.

Table 4-16. Attribute VendorId (CommonPublishedInformation) detailed description

Property	Value
Type	INTEGER_LABEL
Origin	Custom
Symbolic Name	false
Default	43
Invalid	Range >=43 <=43

4.5 Form PortConfigSet

This container contains a configuration of the PORT driver / SIUL module. This container is a MultipleConfigurationContainer, i.e. this container and its sub-containers exit once per configuration set. NOTE:"User should configure only ONE instance of PortConfigSet container and not MULTIPLE."

Included forms :

- [Form UnusedPortPin](#)
- [Form PortContainer](#)

In...	Name	PortNumberOfPortPins
0	GPIO	9
1	ANALOG	2
2	IN_Special	2

Figure 4-4. Tressos Plugin snapshot for PortConfigSet form.

4.5.1 Form UnusedPortPin

Module wide configuration parameters of the PORT driver.

Is included by form : [Form PortConfigSet](#)

CanFilterMaskValue: 2047

Figure 4-5. Tressos Plugin snapshot for UnusedPortPin form.

4.5.1.1 PortPinInitialMode (UnusedPortPin)

Port pin mode from mode list for use with Port_Init() function. NOTE: This parameter is not used in the current implementation and is retained as per std AUTOSAR_EcucParamDef.arxml file.

Table 4-17. Attribute PortPinInitialMode (UnusedPortPin) detailed description

Property	Value
Label	PortPin Initial Mode
Type	ENUMERATION
Origin	Custom
Symbolic Name	false
Default	PORT_GPIO_MODE
Enable	false
Range	PORT_GPIO_MODE PORT_ALT1_FUNC_MODE PORT_ALT2_FUNC_MODE PORT_ALT3_FUNC_MODE PORT_ANALOG_INPUT_MODE PORT_ONLY_INPUT_MODE PORT_OSC_SIGNAL_MODE PORT_EXTRA_INPUT_MODE PORT_EXTRA2_INPUT_MODE

4.5.1.2 PortPinWpe (UnusedPortPin)

Enable Weak Pull Up/Down for the configured Pin. Checked box means any of the Weak Pull Up or Weak Pull Down configurations is enabled for the pin. This parameter is not supported for the pins configured as analog pin. This is an implementation specific parameter.

Table 4-18. Attribute PortPinWpe (UnusedPortPin) detailed description

Property	Value
Label	PortPin Wpe
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.5.1.3 PortPinReadback (UnusedPortPin)

Enable Readback capability for the configured Pin. This is an implementation specific parameter.

Table 4-19. Attribute PortPinReadback (UnusedPortPin) detailed description

Property	Value
Label	PortPin Readback
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.5.1.4 PortPinSafeMode (UnusedPortPin)

Enable/Disable Safe Mode for the pin. Checked box means the Safe Mode is enabled.

Table 4-20. Attribute PortPinSafeMode (UnusedPortPin) detailed description

Property	Value
Label	PortPin Safe Mode
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.5.1.5 PortPinWps (UnusedPortPin)

Select Weak Pull Up/Down for the configured Pin. Checked box means the Weak Pull Up configuration is set. Unchecked box means the Weak Pull Down configuration is set. This parameter is not supported for the pins configured as analog pin. This is an implementation specific parameter.

Table 4-21. Attribute PortPinWps (UnusedPortPin) detailed description

Property	Value
Label	PortPin Wps

Table continues on the next page...

Table 4-21. Attribute PortPinWps (UnusedPortPin) detailed description (continued)

Property	Value
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.5.1.6 PortPinDirection (UnusedPortPin)

Configure DIRECTION for the configured Pin.

Table 4-22. Attribute PortPinDirection (UnusedPortPin) detailed description

Property	Value
Label	PortPin Direction
Type	ENUMERATION
Origin	Custom
Symbolic Name	false
Default	PORT_PIN_DISABLED
Range	PORT_PIN_IN PORT_PIN_OUT PORT_PIN_DISABLED

4.5.1.7 PortPinLevelValue (UnusedPortPin)

Port Pin Level value from Port pin list.

Table 4-23. Attribute PortPinLevelValue (UnusedPortPin) detailed description

Property	Value
Label	PortPin Level Value
Type	ENUMERATION
Origin	Custom
Symbolic Name	false
Default	PORT_PIN_LEVEL_LOW
Range	PORT_PIN_LEVEL_HIGH PORT_PIN_LEVEL_LOW

4.5.2 Form PortContainer

Container collecting the PortPins.

Is included by form : [Form PortConfigSet](#)

Included forms :

- [Form PortPin](#)

Figure 4-6. Tressos Plugin snapshot for PortContainer form.

4.5.2.1 PortNumberOfPortPins (PortContainer)

The number of specified PortPins in this PortContainer.

Table 4-24. Attribute PortNumberOfPortPins (PortContainer) detailed description

Property	Value
Label	PortNumberOfPortPins
Type	INTEGER
Origin	AUTOSAR_ECUC
Symbolic Name	false
Invalid	Range >=1 <=500

4.5.2.2 Form PortPin

Configuration of the individual port pins.

Is included by form : [Form PortContainer](#)

Figure 4-7. Tresos Plugin snapshot for PortPin form.

4.5.2.2.1 PortPinWpe (PortPin)

Enable Weak Pull Up/Down for the configured Pin. Checked box means any of the Weak Pull Up or Weak Pull Down configurations is enabled for the pin. This is an implementation specific parameter.

Table 4-25. Attribute PortPinWpe (PortPin) detailed description

Property	Value
Label	PortPin Wpe
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.5.2.2.2 PortPinWps (PortPin)

Select Weak Pull Up/Down for the configured Pin. Checked box means the Weak Pull Up configuration is set. Unchecked box means the Weak Pull Down configuration is set. This is an implementation specific parameter.

Table 4-26. Attribute PortPinWps (PortPin) detailed description

Property	Value
Label	PortPin Wps
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.5.2.2.3 PortPinOde (PortPin)

Enable Open Drain Output for the configured Pin. Checked box means the Open Drain configuration is set. This is an implementation specific parameter.

Table 4-27. Attribute PortPinOde (PortPin) detailed description

Property	Value
Label	PortPin Ode
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.5.2.2.4 PortPinSafeMode (PortPin)

Enable/Disable Safe Mode for the pin. Checked box means the Safe Mode is enabled.

Table 4-28. Attribute PortPinSafeMode (PortPin) detailed description

Property	Value
Label	PortPin Safe Mode
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.5.2.2.5 PortPinDirectionChangeable (PortPin)

Enable/Disable the changeability for the configured Pin. Checked box means the Direction Changeability is enabled. This is an implementation specific parameter.

Table 4-29. Attribute PortPinDirectionChangeable (PortPin) detailed description

Property	Value
Label	PortPin Direction Changeable
Type	BOOLEAN
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	true

4.5.2.2.6 PortPinReadback (PortPin)

Enable Readback capability for the configured Pin. This is an implementation specific parameter.

Table 4-30. Attribute PortPinReadback (PortPin) detailed description

Property	Value
Label	PortPin Readback
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.5.2.2.7 PortPinModeChangeable (PortPin)

Parameter to indicate if the mode of a port pin is changeable during runtime. True: Port Pin mode changeable allowed. False: Port Pin mode changeable not permitted

Table 4-31. Attribute PortPinModeChangeable (PortPin) detailed description

Property	Value
Label	PortPin Mode Changeable
Type	BOOLEAN
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	true

4.5.2.2.8 PortPinId (PortPin)

Pin Id of the port pin. This value will be assigned to the symbolic name derived from the port pin container short name.

Table 4-32. Attribute PortPinId (PortPin) detailed description

Property	Value
Label	PortPin Id
Type	INTEGER
Origin	AUTOSAR_ECUC
Symbolic Name	true
Invalid	Range >=1 <=256

4.5.2.2.9 PortPinPcr (PortPin)

Pin Id of the port pin. This value will be assigned to the symbolic name derived from the port pin container short name.

Table 4-33. Attribute PortPinPcr (PortPin) detailed description

Property	Value
Label	PortPinPcr
Type	INTEGER
Origin	Custom
Symbolic Name	false
Invalid	Range >=0 <=255

4.5.2.2.10 PortPinDirection (PortPin)

Selects the initial direction of the pin (IN or OUT). If the direction is not changeable, the value configured here is fixed. The pin direction can be set only for the GPIO pins. For the Alternative Function modes the OUT pin direction is hw selected. If the IN direction is needed too, it can be set at runtime. NOTE: To set the IN direction take care, please, that all the possible module inputs, possible as Alternative Functions for the pad mode, are hw connected together, if IN direction is enabled, to the pad.

Table 4-34. Attribute PortPinDirection (PortPin) detailed description

Property	Value
Label	PortPin Direction
Type	ENUMERATION
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	PORT_PIN_IN
Range	PORT_PIN_IN PORT_PIN_OUT

4.5.2.2.11 PortPinInitialMode (PortPin)

Port pin mode from mode list for use with Port_Init() function. NOTE: This parameter is not used in the current implementation and is retained as per std AUTOSAR_EcucParamDef.arxml file.

Table 4-35. Attribute PortPinInitialMode (PortPin) detailed description

Property	Value
Label	PortPin Initial Mode
Type	ENUMERATION
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	PORT_GPIO_MODE
Enable	false
Range	PORT_GPIO_MODE PORT_ALT1_FUNC_MODE PORT_ALT2_FUNC_MODE PORT_ALT3_FUNC_MODE PORT_ANALOG_INPUT_MODE PORT_ONLY_INPUT_MODE PORT_OSC_SIGNAL_MODE PORT_EXTRA_INPUT_MODE PORT_EXTRA2_INPUT_MODE

4.5.2.2.12 PortPinMode (PortPin)

Selects the PORT pin mode from the modes list. By default more than one mode are allowed. That way it is e.g. possible to combine DIO with another mode such as ICU. For the Alternative Function modes (not a GPIO mode) the OUT direction is hw selected for that pin. If the IN direction is needed too, it can be set at runtime. NOTE: To set the IN

direction take care, please, that all the possible module inputs, possible as Alternative Functions for the pad mode, are hw connected together, if IN direction is enabled, to the pad.

Table 4-36. Attribute PortPinMode (PortPin) detailed description

Property	Value
Label	PortPin Mode
Type	ENUMERATION
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	GPIO

4.5.2.2.13 PortPinLevelValue (PortPin)

Port Pin Level value from Port pin list.

Table 4-37. Attribute PortPinLevelValue (PortPin) detailed description

Property	Value
Label	PortPin Level Value
Type	ENUMERATION
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	PORT_PIN_LEVEL_LOW
Range	PORT_PIN_LEVEL_HIGH PORT_PIN_LEVEL_LOW

4.5.2.2.14 PortPinSlewRate (PortPin)

Configure Slew Rate for the configured Pin. This is an implementation specific parameter.

Table 4-38. Attribute PortPinSlewRate (PortPin) detailed description

Property	Value
Label	PortPin Output Slew Rate
Type	ENUMERATION
Origin	Custom
Symbolic Name	false
Default	SlowestConfiguration
Range	SlowestConfiguration FastestConfiguration

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